

Homework One

Problem One

part a

```
void F(int n) {
    int i = 2;
    while (i < n) {
        i = i * i;
    }
}
```

take $\log_2$ both sides
 $\log(2^+) \geq \log_2 n$
 take $\log$ again
 $+ \geq \log_2(\log_2 n)$
 Runtime: $\Theta(\log(\log n))$

$2^{2^+} \geq n$

part b:

```
void f2C(int n) {
    for (int i = 1; i <= n; i++) {
        if ((i % (int) sqrt(n)) == 0) {
            for (int k = 0; k < pow(i, 2); k++) {
                // something O(1) * /
            }
        }
    }
}
```

$\sum_{i=1}^n \Theta(1) + \sum_{i=1}^n i^2 n^{3/2}$
 $n^{3/2} \sum_{i=1}^n i^2$
 $= \Theta(n^4)$
 $(\sqrt{n})^4 = n^2$
 $n^{3/2} \cdot \Theta(n^2)$
 $n^{3/2} \cdot n^2 = n^{7/2}$
 $\Theta(n^{7/2})$

part c:

```
for (int i = 1; i <= n; i++) {
    for (int u = 1; u <= n; u++) {
        if (A[u] == i) {
            for (int m = 1; m <= n; m = m + m) {
                // something O(1) * /
            }
        }
    }
}
```

$\Theta(n^2)$
 $\Theta(n^2)$
 $\Theta(n^2)$
 $A[u] = i$, only true once max
 $n \cdot \Theta(\log n) = \Theta(n \log n)$
 $\Theta(n^2 + n \log n)$
 $2^+ > n$ $\log_2 \Theta(\log n)$
 this grows faster
Answer: $\Theta(n^2)$

part d

```
int F(int n) {
    int *a = new int[10];
    int size = 10;
    for (int i = 0; i < n; i++) {
        if (i == size) {
            int newSize = 3 * size / 2;
            int *b = new int[newSize];
            for (int j = 0; j < size; j++) b[j] = a[j];
            delete[] a;
            a = b;
            size = newSize;
        }
        a[i] = i * i;
    }
}
```

$\Theta(n)$
 geometric
 $\Theta(n) + \Theta(n)$
 $= \Theta(2n) = \Theta(n)$

Homework One

Problem Two

Question a:

Initially:

in1: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \text{nullptr}$

in2: $5 \rightarrow 6 \rightarrow \text{nullptr}$

$\text{in1} \rightarrow \text{next} = \text{llrec}(\text{in2}, \text{in1} \rightarrow \text{next})$

return in1

$\text{p1} \rightarrow \text{next} = \text{llrec}(\text{p5}, \text{p2})$

return p1

$\text{llrec}(\text{p1}, \text{p5})$

$\hookrightarrow \text{p1} \rightarrow \text{next} = \text{llrec}(\text{p5}, \text{p2})$

$\hookrightarrow \text{llrec}(\text{p5}, \text{p2})$

$\hookrightarrow \text{p5} \rightarrow \text{next} = \text{llrec}(\text{p2}, \text{p6})$

$\hookrightarrow \text{llrec}(\text{p2}, \text{p6})$

$\hookrightarrow \text{p2} \rightarrow \text{next} = \text{llrec}(\text{p6}, \text{p5})$

$\hookrightarrow \text{llrec}(\text{p6}, \text{p5})$

$\hookrightarrow \text{p6} \rightarrow \text{next} = \text{llrec}(\text{p5}, \text{nullptr})$

$\hookrightarrow \text{llrec}(\text{p5}, \text{nullptr})$
return p3

substitute

$\text{p5} \rightarrow \text{next} = \text{llrec}(\text{p2}, \text{p6})$

$\text{p5} \rightarrow \text{next}$ goes to p6

$\text{p2} \rightarrow \text{next} = \text{p6}$

$\text{p2} \rightarrow \text{p6} \rightarrow \text{p3} \rightarrow \text{p4} \rightarrow \text{nullptr}$

$\text{p5} \rightarrow \text{next} = \text{p2}$

$\text{p5} \rightarrow \text{p2} \rightarrow \text{p3} \rightarrow \text{p4} \rightarrow \text{nullptr}$

$\text{p1} \rightarrow \text{next} = \text{p1}$

$\text{p1} \rightarrow \text{p5} \rightarrow \text{p2} \rightarrow \text{p3} \rightarrow \text{p4} \rightarrow \text{nullptr}$

trace $\text{llrec}(\text{p2}, \text{p6})$

$\text{p2} \rightarrow \text{next} = \text{llrec}(\text{p6}, \text{p3})$

return p2

$\text{p2} \rightarrow \text{next}$ goes to p3

$\text{llrec}(\text{p6}, \text{p3})$

$\text{p6} \rightarrow \text{next} = \text{llrec}(\text{p3}, \text{nullptr})$

$\text{in2} == \text{nullptr}$

$\hookrightarrow \text{return in1} = \text{p3}$

$\text{p6} \rightarrow \text{next} = \text{p3}$

$\text{p6} \rightarrow \text{p3} \rightarrow \text{p4} \rightarrow \text{nullptr}$

remove p5

Answer:

$1 \rightarrow 5 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 4 \rightarrow \text{nullptr}$

Question B:

$\text{llrec}(\text{nullptr}, \text{p2})$

$\hookrightarrow$ if $\text{in1} == \text{nullptr}$ gets called statement

if $(\text{in1} == \text{nullptr}) \{$

return in2

$\}$

$\text{in2} = 2$

returned list
called is
 $2 \rightarrow \text{nullptr}$